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Ahmed Fouad Lagha

AI Engineer / Researcher

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Summary

AI Engineer and PhD candidate with hands-on experience developing and operationalizing end-to-end AI solutions across classical ML, deep learning, and generative AI — from data preparation to production deployment of RAG pipelines, LLM-powered applications, and model inference systems.

Technical Skills

Programming: Python, SQL, C/C++, Java, JavaScript, Bash
ML & AI: PyTorch, Scikit-learn, Pandas, NumPy, OpenCV
GenAI & LLMs: RAG, Vector DBs, Prompt Engineering
Databases: MySQL, PostgreSQL, SQLite, MongoDB
DevOps & Tools: Docker, FastAPI, REST APIs, Git, CI/CD

Experience

AI Research Assistant — **Sep. 2025 – Present**
Eötvös Loránd University (ELTE) — *Budapest, Hungary*

- Develop DP-SGD federated learning pipelines for privacy-preserving synthetic data generation ($\varepsilon = 1$), engineering data preparation and feature workflows to benchmark privacy-utility trade-offs
- Author open-source IDaSec Practicum (8 labs), documenting AI architectures and best practices in responsible AI
- Collaborate with cross-functional research teams and mentor graduate students in AI engineering and MLOps workflows

Data Analyst — **Dec. 2024 – Mar. 2025**
Djezzy — *Khenchela, Algeria*

- Analyzed subscriber telecom datasets using Python and SQL to identify usage trends and customer behavior patterns
- Built automated data preparation and feature engineering pipelines, facilitating data-driven decision making and reporting
- Collaborated with analytics and cross-functional teams to model subscriber metrics and optimize customer retention strategies

Projects

MyVellum — **Dec. 2025 – Present**
FastAPI, Groq API, LLM Orchestration, RAG, Docker, CI/CD

- Architected an LLM-powered app with modular prompt engineering and AI workflow orchestration
- Developed vector-backed RAG pipelines with context-stitching and retrieval optimization, reducing hallucination rates
- Built automated evaluation gates into CI/CD for relevance and quality metrics; integrated Groq API and FastAPI with Docker

Tabular Polygraph — **Jan. 2026 – Present**
Python, PyTorch, Scikit-learn, Pandas, Synthetic Data Auditing

- Engineered an open-source neuro-symbolic framework for evaluating synthetic tabular data fidelity, logical integrity, and privacy
- Developed Python CLI and API tools with automated 4-pillar scorecard diagnostics (Fidelity, Logic, Utility, Privacy)
- Designed statistical metric pipelines and row-level hallucination detection algorithms for production AI/ML validation

Education

PhD in Informatics — Privacy-Preserving Machine Learning — **Sep. 2025 – Expected 2029**
Eötvös Loránd University (ELTE) — *Budapest, Hungary*
Thesis: Synthetic data generation in artificial intelligence security (Supervised by Pr. Lendák Imre)

M.Sc. in Artificial Intelligence (GPA: 3.7/4.0) — **Sep. 2023 – Jun. 2025**
University of Khenchela — *Khenchela, Algeria*
Thesis: Hair Dermoscopic Image Segmentation Using Deep Learning (Supervised by Pr. Dalal Bardou)

B.Sc. in Mathematics and Computer Science (GPA: 3.9/4.0) — **Sep. 2020 – Jun. 2023**
University of Khenchela — *Khenchela, Algeria*

Awards & Certifications

- **Stipendium Hungaricum Scholarship** — Fully funded PhD studies at ELTE — Sep. 2025
- **2nd Place — Starthon'25 Hackathon** — University of El Oued, Algeria — Apr. 2025
- **Software Engineering (Back-end Specialization)** — ALX Africa — Feb. 2025
- **AI Career Essentials (AiCE)** — ALX Africa — Jul. 2024

Languages

Arabic (Native) | English (Professional Proficiency) | French (Intermediate) | Hungarian (Beginner)